

Assignment - 3

① a) $V_T = \frac{kT}{q}$

$$V_T = \frac{1.38 \times 10^{-23} \text{ J/K} \times (273 + 25) \text{ K}}{1.602 \times 10^{-19}}$$

$$V_T = 256.70 \times 10^{-4}$$

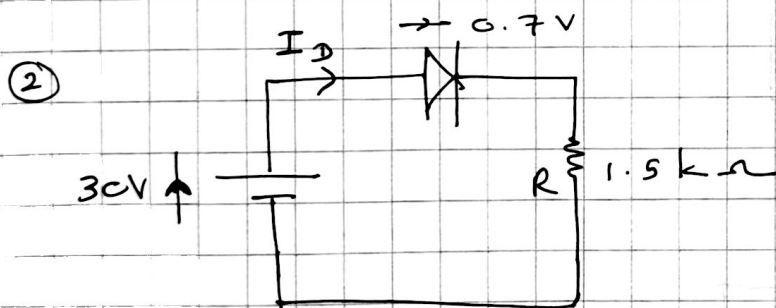
$$V_T = 0.0256 \text{ V} //$$

b) $I_D = I_S (e^{V_D/nV_T} - 1)$

$$I_D = (40 \times 10^{-9}) (e^{0.5/2(0.0256)} - 1)$$

~~$$I_D = 0.364 \text{ A} //$$~~

$$I_D = 0.696 \text{ mA} //$$



$$V_D = 0.7 \text{ V (s. diode knee voltage)}$$

Apply KVL f

$$-30 + 0.7 + 1.5 \times 10^3 I_D = 0$$

$$I_D = \frac{29.3}{1.5 \times 10^3}$$

$$I_D = 0.019 \text{ A} //$$

$$V_R = I_D \times R = 0.019 \times 1.5 \times 10^3 = 28.5 \text{ V} //$$

b) $V_D = 0 \text{ V}$ (No knee voltage in Ideal diode)

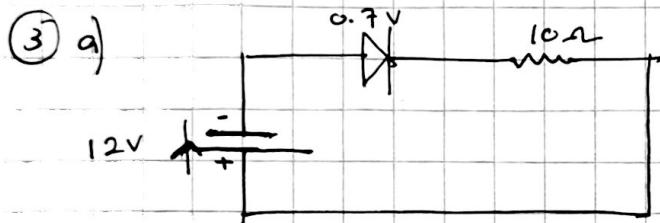
KVL to circuit

$$-30 + I_D (1.5 \times 10^3) = 0$$

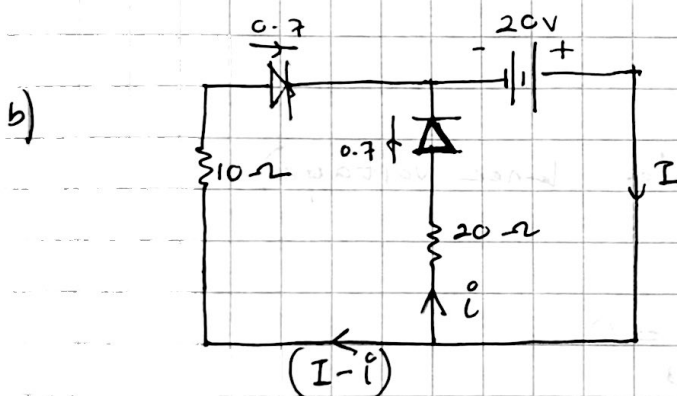
$$I_D = 20 \text{ mA} //$$

$$\begin{aligned} V_R &= I_D \times R \\ &= 20 \text{ mA} \times 1.5 \text{ k}\Omega \\ &= 30 \text{ V} // \end{aligned}$$

c) If voltage is high in the circuit, the drop of 0.7 V is very low affected. So ideal diode or not is not much considerable in high voltage circuits. But making the ideal model a quick and effective tool for estimation.



$$I = 0 \text{ A}$$



KVL

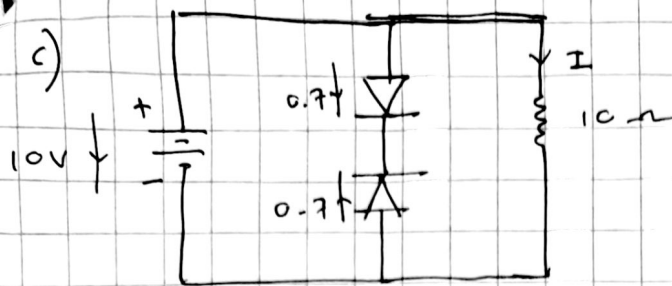
$$\begin{aligned} -20 + (I-i)10 + 0.7 &= 0 \\ 10I - 10i &= 19.3 \rightarrow \textcircled{1} \end{aligned}$$

$$\begin{aligned} 10I - 10(0.965) &= 19.3 \\ I &= 0.965 \text{ A} // \end{aligned}$$

KVL

$$\begin{aligned} -20 + 20i + 0.7 &= 0 \\ 20i &= 19.3 \end{aligned}$$

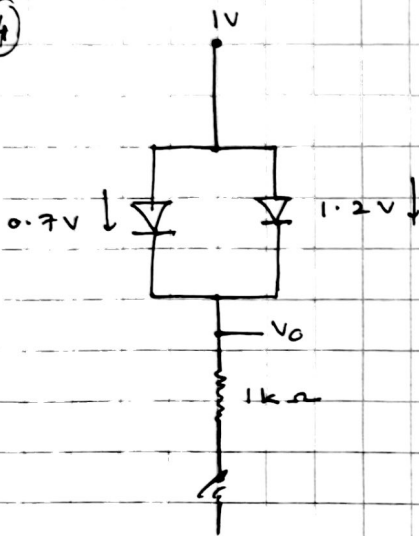
$$i = 0.965 \text{ A}$$



$$10 = 10 I$$

$$I = 1 A //$$

④



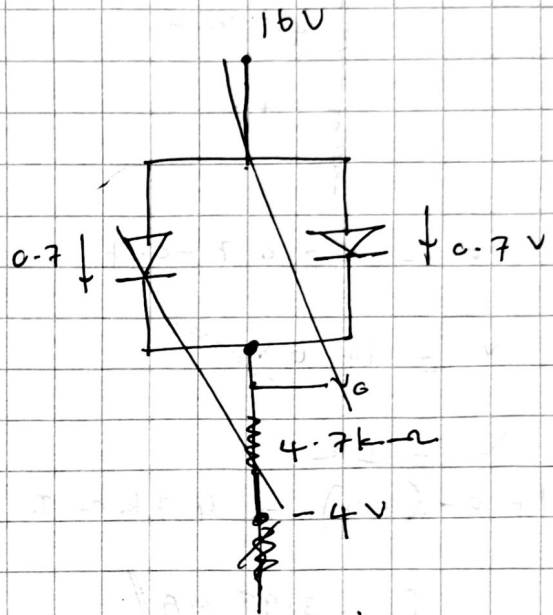
$$V_0 = 1 - 0.7$$

$$= 0.3V$$

$$V = IR$$

$$0.3 = 1k\Omega I$$

$$I = 0.3mA //$$



$$V_0 = 16 - 0.7 - 0.7$$

$$= 15.3V \quad 14.6V$$

$$V = IR$$

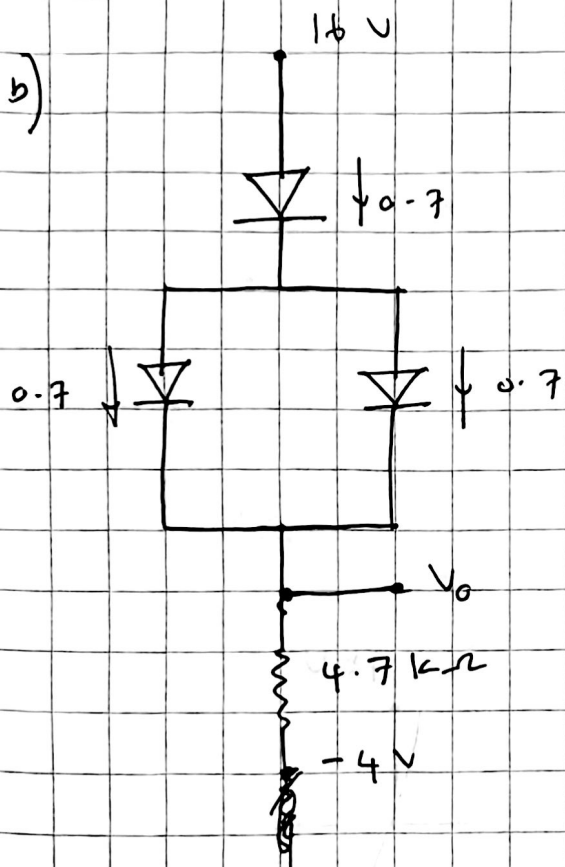
$$14.6 - (-4) = I (4.7k\Omega)$$

$$I = \frac{19.6}{4.7} = \frac{18.6}{4.7k\Omega}$$

$$I = 4.10mA //$$

$$I = 3.95mA //$$

b)



$$V_o = 16 - 0.7 - 0.7$$

$$V_o = 14.6 \text{ V}$$

$$V = IR$$

$$14.6 - (-4) = 4.7 \text{ k}\Omega I$$

$$I = 3.95 \text{ mA}$$